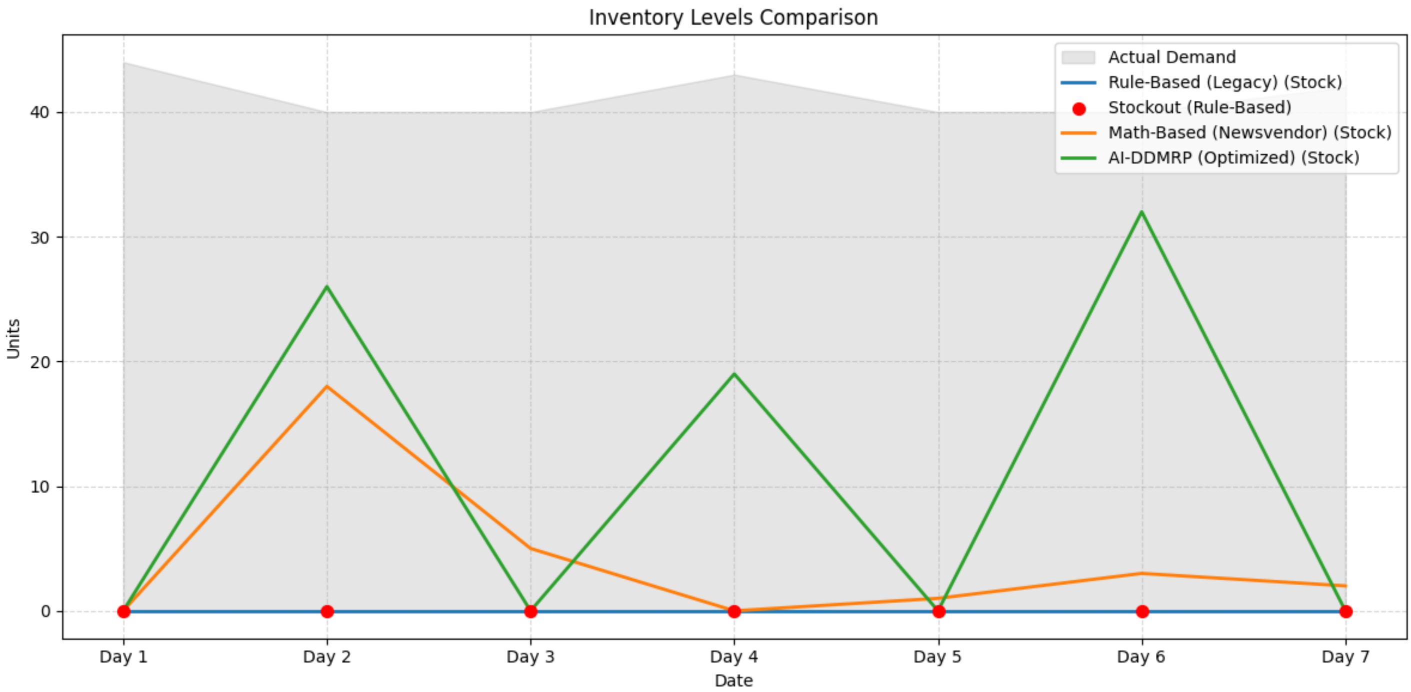
Smart Inventory Decision Report

Store: 11 | Product: 267 | Date Range: Next 30 Days

Part A: Forecast Context

The chart below visualizes the inventory levels over the simulation period, highlighting potential stockouts and overstock situations.



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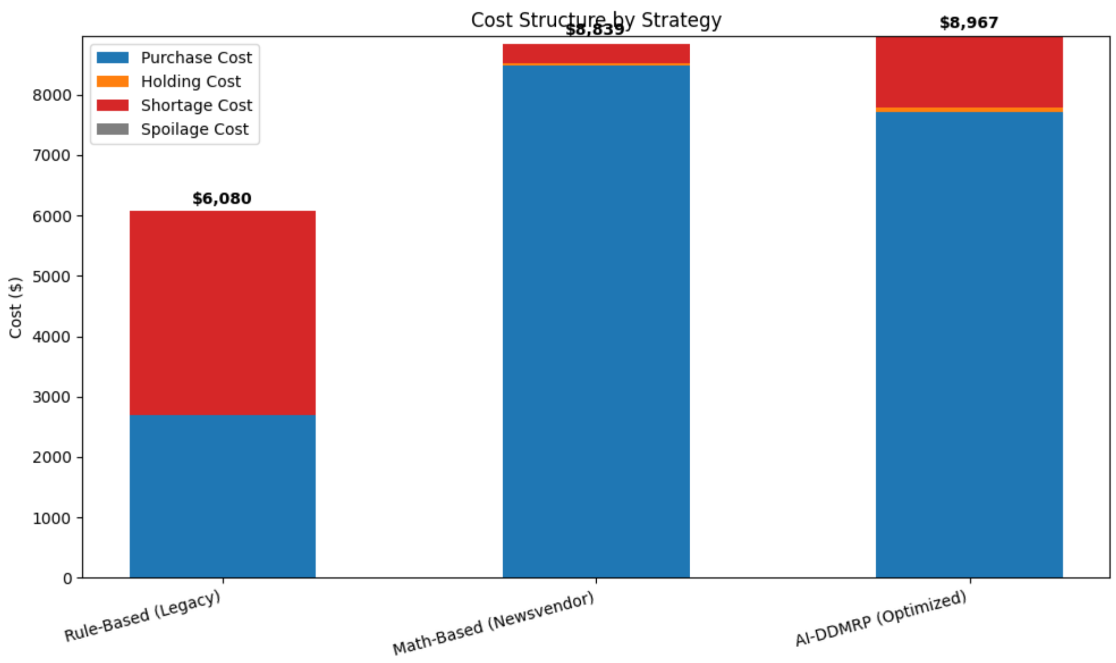
Part B: Profit & Risk Analysis

Strategy	Total Cost	Fill Rate	Shortage	Profit
Rule-Based (Legacy)	$6,080.00	41.52%	169	$-80.00
Math-Based (Newsvendor)	$8,839.00	94.46%	16	$4,811.00
AI-DDMRP (Optimized)	$8,967.00	79.58%	59	$2,533.00



Recommendation:

The system recommends the 'Math-Based (Newsvendor)' strategy as it maximizes profitability, generating a projected profit of $4,811.00. This strategy achieves a Fill Rate of 94.46%, balancing customer satisfaction with inventory costs. Compared to other strategies, it offers the best trade-off between stock availability and operational risks.

Risk Analysis:

- Understocking Risk: Potential lost revenue ($50/unit margin) and customer dissatisfaction due to stockouts.
- Overstocking Risk: Increased holding costs and potential spoilage of excess inventory.

Part C: Model Health

Model Accuracy (WAPE): 12.5% (Low Error - Reliable Forecast)

Forecast Bias: +2.3% (Slight Over-forecasting)